

Concept 2 | Geometric Sequences

English Student Edition • Focused formulas and guided practice

Formula opener

Term

$$a_n = ar^{n-1}$$

Finite sum

$$S_n = a \frac{r^n - 1}{r - 1}$$

Infinite sum

$$S_{\infty} = \frac{a}{1-r} \quad (|r| < 1)$$

Classified Practice

Geometric Sequences

Q001 **Written**

Find the general term of a geometric sequence whose first term is 2 and common ratio is 3.

Q002 Written

Find the general term when the first term is 3 and the common ratio is 4.

Classified Practice

Geometric Sequences

Q003 Written

Find the general term of the geometric sequence in part a. The first term is 5, common ratio 2.

Q004 Written

Find the general term of the geometric sequence in part b. The first term is 2, common ratio 5.

Classified Practice

Geometric Sequences

Q005 **Written**

Find the general term of the geometric sequence in part c. The first term is 7, common ratio 4.

Q006 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = 3$, and $n = 5$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 2(3)^4$

B $a_n = 2(3)^5$

C $a_n = 2(3)^3$

D $a_n = 2 + (3)^4$

Classified Practice

Geometric Sequences

Q007 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = 2$, and $n = 6$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 3(2)^5$

B $a_n = 3(2)^6$

C $a_n = 3(2)^4$

D $a_n = 3 + (2)^5$

Q008 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 4$, and $n = 4$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 5(4)^3$

B $a_n = 5(4)^4$

C $a_n = 5(4)^2$

D $a_n = 5 + (4)^3$

Classified Practice

Geometric Sequences

Q009 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = 5$, and $n = 3$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 4(5)^2$

B $a_n = 4(5)^3$

C $a_n = 4(5)^1$

D $a_n = 4 + (5)^2$

Q010 MCQ

Select the correct option: For first term $a = 5$, common ratio $r = 2$, and $n = 7$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 5(2)^6$

B $a_n = 5(2)^7$

C $a_n = 5(2)^5$

D $a_n = 5 + (2)^6$

Classified Practice

Geometric Sequences

Q011 MCQ

Select the correct option: For first term $a = 7$, common ratio $r = 4$, and $n = 3$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 7(4)^2$

B $a_n = 7(4)^3$

C $a_n = 7(4)^1$

D $a_n = 7 + (4)^2$

Q012 Written

Find the general term when the first term is 5 and the common ratio is -2 .

Classified Practice

Geometric Sequences

Q013 Written

Find the general term of the geometric sequence in part d. The first term is 2, common ratio -3 .

Q014 Written

Find the general term of the geometric sequence in part e. The first term is 3, common ratio -5 .

Classified Practice

Geometric Sequences

Q015 **Written**

Find the general term of the geometric sequence in part f. The first term is 4, common ratio -2 .

Q016 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = -3$, and $n = 5$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 2(-3)^4$

B $a_n = 2(-3)^5$

C $a_n = 2(-3)^3$

D $a_n = 2 + (-3)^4$

Classified Practice

Geometric Sequences

Q017 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -2$, and $n = 6$, which expression gives a_n ?

Choose the correct option.

A $a_n = 3(-2)^5$

B $a_n = 3(-2)^6$

C $a_n = 3(-2)^4$

D $a_n = 3 + (-2)^5$

Q018 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = -1$, and $n = 4$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 6(-1)^3$

B $a_n = 6(-1)^4$

C $a_n = 6(-1)^2$

D $a_n = 6 + (-1)^3$

Classified Practice

Geometric Sequences

Q019 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -2$, and $n = 3$, which expression gives a_n ?

Choose the correct option.

A $a_n = 8(-2)^2$

B $a_n = 8(-2)^3$

C $a_n = 8(-2)^1$

D $a_n = 8 + (-2)^2$

Q020 MCQ

Select the correct option: For first term $a = 9$, common ratio $r = -2$, and $n = 5$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 9(-2)^4$

B $a_n = 9(-2)^5$

C $a_n = 9(-2)^3$

D $a_n = 9 + (-2)^4$

Classified Practice

Geometric Sequences

Q021 Written

Find the general term when the first term is 8 and the common ratio is $-\frac{1}{3}$.

Q022 Written

Find the general term of the geometric sequence in part g. The first term is 4, common ratio $\frac{1}{3}$.

Classified Practice

Geometric Sequences

Q023 Written

Find the general term of the geometric sequence in part h. The first term is 3, common ratio $-\frac{1}{2}$.

Q024 Written

Find the general term of the geometric sequence in part i. The first term is 6, common ratio $\frac{1}{2}$.

Classified Practice

Geometric Sequences

Q025 MCQ

Select the correct option: For first term $a = 4$, common ratio $r = \frac{1}{3}$, and $n = 5$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 4\left(\frac{1}{3}\right)^4$

B $a_n = 4\left(\frac{1}{3}\right)^5$

C $a_n = 4\left(\frac{1}{3}\right)^3$

D $a_n = 4 + \left(\frac{1}{3}\right)^4$

Q026 MCQ

Select the correct option: For first term $a = 3$, common ratio $r = -\frac{1}{2}$, and $n = 6$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 3\left(-\frac{1}{2}\right)^5$

B $a_n = 3\left(-\frac{1}{2}\right)^6$

C $a_n = 3\left(-\frac{1}{2}\right)^4$

D $a_n = 3 + \left(-\frac{1}{2}\right)^5$

Classified Practice

Geometric Sequences

Q027 MCQ

Select the correct option: For first term $a = 6$, common ratio $r = \frac{1}{2}$, and $n = 4$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 6 \left(\frac{1}{2}\right)^3$

B $a_n = 6 \left(\frac{1}{2}\right)^4$

C $a_n = 6\left(\frac{1}{2}\right)^2$

D $a_n = 6 + \left(\frac{1}{2}\right)^3$

Q028 MCQ

Select the correct option: For first term $a = 2$, common ratio $r = \frac{3}{2}$, and $n = 5$, which expression gives a_n ?

Choose the correct option.

A $a_n = 2 \left(\frac{3}{2}\right)^4$

B $a_n = 2 \left(\frac{3}{2}\right)^5$

C $a_n = 2 \left(\frac{3}{2}\right)^3$

D $a_n = 2 + \left(\frac{3}{2}\right)^4$

Classified Practice

Geometric Sequences

Q029 MCQ

Select the correct option: For first term $a = 8$, common ratio $r = -\frac{3}{2}$, and $n = 4$, which expression gives a_n ?:

Choose the correct option.

A $a_n = 8 \left(-\frac{3}{2}\right)^3$

B $a_n = 8 \left(-\frac{3}{2}\right)^4$

C $a_n = 8 \left(-\frac{3}{2}\right)^2$

D $a_n = 8 + \left(-\frac{3}{2}\right)^3$

Q030 Written**Find the general term of 5, 10, 20, 40,**

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Geometric Sequences

Q031 **Written****Find the general term of 2, 6, 18, 54,**

Q032 Written

Find the general term of the geometric sequence in part j. The sequence is 3, 6, 12, 24,

Classified Practice

Geometric Sequences

Q033 MCQ

Select the correct option: Find the general term of the geometric sequence 3, 6, 12, 24, ...:**Choose the correct option.**

A $3 \times 2^{n-1}$

B 3×2^n

C $3 \times 2^{n-2}$

D 0

Q034 MCQ

Select the correct option: Find the general term of the geometric sequence 2, 10, 50, 250, ...:**Choose the correct option.**

A $2 \times 5^{n-1}$

B 2×5^n

C $2 \times 5^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q035 MCQ

Select the correct option: Find the general term of the geometric sequence 4, 8, 16, 32, ...:**Choose the correct option.**

A $4 \times 2^{n-1}$

B 4×2^n

C $4 \times 2^{n-2}$

D 0

Q036 MCQ

Select the correct option: Find the general term of the geometric sequence 7, 21, 63, 189, ...:**Choose the correct option.**

A $7 \times 3^{n-1}$

B 7×3^n

C $7 \times 3^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q037 MCQ

Select the correct option: Find the general term of the geometric sequence 1, 4, 16, 64, ...:

Choose the correct option.

A 4^{n-1}

B 4^n

C 4^{n-2}

D 0

Q038 Written

Find the general term of the geometric sequence in part k. The sequence is $2, -6, 18, -54, \dots$

Classified Practice

Geometric Sequences

Q039 Written

Find the general term of the geometric sequence in part I. The sequence is $5, -10, 20, -40, \dots$

Q040 MCQ**Select the correct option: Find the general term of the geometric sequence $5, -10, 20, -40, \dots$:****Choose the correct option.**

A $5(-2)^{n-1}$

B $5(-2)^n$

C $5(-2)^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q041 MCQ

Select the correct option: Find the general term of the geometric sequence $2, -6, 18, -54, \dots$:**Choose the correct option.**

A $2(-3)^{n-1}$

B $2(-3)^n$

C $2(-3)^{n-2}$

D 0

Q042 MCQ

Select the correct option: Find the general term of the geometric sequence $-4, 8, -16, 32, \dots$:**Choose the correct option.**

A $-4(-2)^{n-1}$

B $-4(-2)^n$

C $-4(-2)^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q043 MCQ

Select the correct option: Find the general term of the geometric sequence $6, -18, 54, -162, \dots$:**Choose the correct option.**

A $6(-3)^{n-1}$

B $6(-3)^n$

C $6(-3)^{n-2}$

D 0

Q044 Written**Find the general term of $-4, 6, -9, \frac{27}{2}, \dots$**

Classified Practice

Geometric Sequences

Q045 **Written****Find the general term of $-8, 12, -18, 27, \dots$**

Q046 Written**Find the general term of $2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \dots$**

Classified Practice

Geometric Sequences

Q047 Written

Find the general term of the geometric sequence in part m. The sequence is 48, 24, 12, 6,

Q048 Written

Find the general term of the geometric sequence in part n. The sequence is $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \frac{9}{16}, \dots$

Classified Practice

Geometric Sequences

Q049 Written

Find the general term of the geometric sequence in part o. The sequence is $8, 16\sqrt{2}, 64, 128\sqrt{2}, \dots$

Q050 Written

Find the general term of the geometric sequence in part p. The sequence is $3, -\frac{15}{2}, \frac{75}{4}, -\frac{375}{8}, \dots$

Classified Practice

Geometric Sequences

Q051 Written

Find the general term of the geometric sequence in part q. The sequence is $9, -3\sqrt{3}, 3, -\sqrt{3}, \dots$

Q052 Written

Find the general term of the geometric sequence in part r. The sequence is $5, -\frac{10}{3}, \frac{20}{9}, -\frac{40}{27}, \dots$

Classified Practice

Geometric Sequences

Q053 MCQ

Select the correct option: Find the general term of the geometric sequence $-8, 12, -18, 27, \dots$:**Choose the correct option.**

A $-8 \left(-\frac{3}{2}\right)^{n-1}$

B $-8 \left(-\frac{3}{2}\right)^n$

C $-8 \left(-\frac{3}{2}\right)^{n-2}$

D 0

Q054 MCQ

Select the correct option: Find the general term of the geometric sequence 48, 24, 12, 6, ...:**Choose the correct option.**

A $48 \left(\frac{1}{2}\right)^{n-1}$

B $48 \left(\frac{1}{2}\right)^n$

C $48 \left(\frac{1}{2}\right)^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q055 MCQ

Select the correct option: Find the general term of the geometric sequence $\frac{1}{6}, \frac{1}{4}, \frac{3}{8}, \dots$:

Choose the correct option.

- A $\frac{1}{6} \left(\frac{3}{2}\right)^{n-1}$
B $\frac{1}{6} \left(\frac{3}{2}\right)^n$
C $\frac{1}{6} \left(\frac{3}{2}\right)^{n-2}$
D 0

Q056 MCQ

Select the correct option: Find the general term of the geometric sequence $8, 16\sqrt{2}, 64, \dots$:

Choose the correct option.

A $8(2\sqrt{2})^{n-1}$

B $8(2\sqrt{2})^n$

C $8(2\sqrt{2})^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q057 MCQ

Select the correct option: Find the general term of the geometric sequence $9, -3\sqrt{3}, 3, \dots$:

Choose the correct option.

A $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$

B $9\left(-\frac{\sqrt{3}}{3}\right)^n$

C $9\left(-\frac{\sqrt{3}}{3}\right)^{n-2}$

D 0

Q058 MCQ

Select the correct option: Find the general term of the geometric sequence $3, -\frac{3}{2}, \frac{3}{4}, \dots$:

Choose the correct option.

A $3\left(-\frac{1}{2}\right)^{n-1}$

B $3\left(-\frac{1}{2}\right)^n$

C $3\left(-\frac{1}{2}\right)^{n-2}$

D 0

Classified Practice

Geometric Sequences

Q059 MCQ

Select the correct option: Find the general term of the geometric sequence $10, 5, \frac{5}{2}, \dots$:

Choose the correct option.

A $10\left(\frac{1}{2}\right)^{n-1}$

B $10\left(\frac{1}{2}\right)^n$

C $10\left(\frac{1}{2}\right)^{n-2}$

D 0

Q060 Written**Find the common ratio and missing values in $A, -18, 6, B, \dots$**

Classified Practice

Geometric Sequences

Q061 MCQ

Select the correct option: The sequence $A, 12, 6, B, \dots$ is geometric. Find r, A, B :

Choose the correct option.

A $r = \frac{1}{2}, A = 24, B = 3$

B $r = \frac{2}{3}, A = 1, B = 2$

C $r = -1, A = 0, B = 0$

D $r = 3, A = 3, B = 9$

Q062 Written

Find the common ratio and the missing values in $A, 24, 12, B, \dots$

Classified Practice

Geometric Sequences

Q063 Written

Find the common ratio and missing values in $A, -3, 9, B$.

Q064 Written

Find the common ratio and missing values in $A, \frac{1}{3}, \frac{1}{9}, B$.

Classified Practice

Geometric Sequences

Q065 Written

Find the common ratio and missing values in $A, -\frac{1}{4}, \frac{1}{8}, B$.

Classified Practice

Geometric Sequences

Q066 Challenge

In an increasing geometric sequence, if $a_3 + a_5 = 80$ and $a_4 = 24$, find a_7 .

Classified Practice

Geometric Sequences

Q067 **Written**

Find the general term of a geometric sequence whose 4th term is 24 and 6th term is 96.

Q068 Written**Find the general term when the 3rd term is 18 and the 5th term is 162.**

Classified Practice

Geometric Sequences

Q069 **Written**

Find the requested values. The 2nd term is 6 and the 4th term is 54.

Q070 Written

Find the requested values. The 3rd term is 27 and the 5th term is 243.

Classified Practice

Geometric Sequences

Q071 Written

Find the requested values. The 2nd term is 14 and the 5th term is 112.

Q072 Written**Find the general term when the 3rd term is 12 and the 6th term is -96 .**

Classified Practice

Geometric Sequences

Q073 Written

Find the general term when the 2nd term is -6 and the 5th term is 48 .

Q074 Written**Find the general term when the 3rd term is 12 and the 7th term is 192.**

Classified Practice

Geometric Sequences

Q075 **Written**

Find the requested values. The 3rd term is -28 and the 5th term is -112 .

Q076 Written

Find the requested values. The 4th term is 16 and the 7th term is -128 .

Classified Practice

Geometric Sequences

Q077 Written

Find the requested values. The 3rd term is -20 and the 6th term is 160 .

Q078 Written

Find the requested values. The 2nd term is 3 and the 6th term is 768.

Classified Practice

Geometric Sequences

Q079 Written

Find the requested values. The 3rd term is 9 and the 7th term is $729/16$.

Q080 Written

Find the requested values. The 4th term is $\frac{2}{9}$ and the 8th term is 18.

Classified Practice

Geometric Sequences

Q081 MCQ

Select the correct option: For $a = 2$, $r = 3$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 2(3)^{n-1}$

B $a_n = 2(3)^n$

C $a_n = 2(3)^{n+1}$

D $a_n = 2 + 3^{n-1}$

Q082 MCQ

Select the correct option: For $a = 3$, $r = 2$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 3(2)^{n-1}$

B $a_n = 3(2)^n$

C $a_n = 3(2)^{n+1}$

D $a_n = 3 + 2^{n-1}$

Classified Practice

Geometric Sequences

Q083 MCQ

Select the correct option: For $a = 5$, $r = 4$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 5(4)^{n-1}$

B $a_n = 5(4)^n$

C $a_n = 5(4)^{n+1}$

D $a_n = 5 + 4^{n-1}$

Q084 MCQ

Select the correct option: For $a = 4$, $r = 5$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 4(5)^{n-1}$

B $a_n = 4(5)^n$

C $a_n = 4(5)^{n+1}$

D $a_n = 4 + 5^{n-1}$

Classified Practice

Geometric Sequences

Q085 MCQ

Select the correct option: For $a = 2$, $r = -3$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 2(-3)^{n-1}$

B $a_n = 2(-3)^n$

C $a_n = 2(-3)^{n+1}$

D $a_n = 2 - 3^{n-1}$

Q086 MCQ

Select the correct option: For $a = 3$, $r = -2$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 3(-2)^{n-1}$

B $a_n = 3(-2)^n$

C $a_n = 3(-2)^{n+1}$

D $a_n = 3 - 2^{n-1}$

Classified Practice

Geometric Sequences

Q087 MCQ

Select the correct option: For $a = 6$, $r = -1$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 6(-1)^{n-1}$

B $a_n = 6(-1)^n$

C $a_n = 6(-1)^{n+1}$

D $a_n = 6 - 1^{n-1}$

Q088 MCQ

Select the correct option: For $a = 8$, $r = -2$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 8(-2)^{n-1}$

B $a_n = 8(-2)^n$

C $a_n = 8(-2)^{n+1}$

D $a_n = 8 - 2^{n-1}$

Classified Practice

Geometric Sequences

Q089 MCQ

Select the correct option: For $a = 4$, $r = \frac{1}{3}$, which expression gives the general term a_n :

Choose the correct option.

- A $a_n = 4\left(\frac{1}{3}\right)^{n-1}$
- B $a_n = 4\left(\frac{1}{3}\right)^n$
- C $a_n = 4\left(\frac{1}{3}\right)^{n+1}$
- D $a_n = 4 + \frac{1}{3}^{n-1}$

Q090 MCQ

Select the correct option: For $a = 3$, $r = -\frac{1}{2}$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 3 \left(-\frac{1}{2}\right)^{n-1}$

B $a_n = 3 \left(-\frac{1}{2}\right)^n$

C $a_n = 3 \left(-\frac{1}{2}\right)^{n+1}$

D $a_n = 3 - \frac{1}{2}^{n-1}$

Classified Practice

Geometric Sequences

Q091 MCQ

Select the correct option: For $a = 6$, $r = \frac{1}{2}$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 6 \left(\frac{1}{2}\right)^{n-1}$

B $a_n = 6 \left(\frac{1}{2}\right)^n$

C $a_n = 6 \left(\frac{1}{2}\right)^{n+1}$

D $a_n = 6 + \frac{1}{2}^{n-1}$

Q092 MCQ

Select the correct option: For $a = 2$, $r = \frac{3}{2}$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 2 \left(\frac{3}{2}\right)^{n-1}$

B $a_n = 2 \left(\frac{3}{2}\right)^n$

C $a_n = 2 \left(\frac{3}{2}\right)^{n+1}$

D $a_n = 2 + \frac{3^{n-1}}{2}$

Classified Practice

Geometric Sequences

Q093 MCQ

Select the correct option: For $a = 8$, $r = -\frac{3}{2}$, which expression gives the general term a_n :

Choose the correct option.

- A $a_n = 8 \left(-\frac{3}{2}\right)^{n-1}$
- B $a_n = 8 \left(-\frac{3}{2}\right)^n$
- C $a_n = 8 \left(-\frac{3}{2}\right)^{n+1}$
- D $a_n = 8 - \frac{3^{n-1}}{2}$

Q094 MCQ

Select the correct option: For $a = 5$, $r = 2$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 5(2)^{n-1}$

B $a_n = 5(2)^n$

C $a_n = 5(2)^{n+1}$

D $a_n = 5 + 2^{n-1}$

Classified Practice

Geometric Sequences

Q095 MCQ

Select the correct option: For $a = 7$, $r = 4$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 7(4)^{n-1}$

B $a_n = 7(4)^n$

C $a_n = 7(4)^{n+1}$

D $a_n = 7 + 4^{n-1}$

Q096 MCQ

Select the correct option: For $a = 9$, $r = -2$, which expression gives the general term a_n :

Choose the correct option.

A $a_n = 9(-2)^{n-1}$

B $a_n = 9(-2)^n$

C $a_n = 9(-2)^{n+1}$

D $a_n = 9 - 2^{n-1}$

Classified Practice

Geometric Sequences

Q097 Written

When 2, x , 8 form a geometric sequence in this order, find x .

Q098 Written**Find the requested values. When 1, x , 25 form a geometric sequence, find x .**

Classified Practice

Geometric Sequences

Q099 Written

Find the requested values. When $1/2, x, 8$ form a geometric sequence, find x .

Q100 Written

When 1, x , 12 form a geometric sequence in this order, find x .

Classified Practice

Geometric Sequences

Q101 MCQ

Select the correct option: If x is the middle term of the geometric triple $2, x, 8$, find x :

Choose the correct option.

A ± 4

B $+4$

C -4

D 0

Q102 MCQ**Select the correct option: If x is the middle term of the geometric triple $1, x, 25$, find x :****Choose the correct option.**A ± 5 B $+5$ C -5 D 0

Classified Practice

Geometric Sequences

Q103 MCQ

Select the correct option: If x is the middle term of the geometric triple $\frac{1}{2}, x, 8$, find x :

Choose the correct option.

- A ± 2
- B $+2$
- C -2
- D 0

Q104 MCQ**Select the correct option: If x is the middle term of the geometric triple 3, x , 12, find x :****Choose the correct option.**A ± 6 B $+6$ C -6 D 0

Classified Practice

Geometric Sequences

Q105 MCQ

Select the correct option: If x is the middle term of the geometric triple 4, x , 9, find x :

Choose the correct option.

A ± 6

B $+6$

C -6

D 0

Q106 MCQ**Select the correct option: If x is the middle term of the geometric triple $2, x, 18$, find x :****Choose the correct option.**A ± 6 B $+6$ C -6 D 0

Classified Practice

Geometric Sequences

Q107 Written

The numbers $6, a, b$ form an arithmetic sequence and $a, b, 16$ form a geometric sequence. Find a, b .

Q108 Written

Find the requested values. The numbers $a, b, 2$ form an arithmetic sequence and $a, 2, b$ form a geometric sequence. Find a, b .

Classified Practice

Geometric Sequences

Q109 Written

The numbers $-2, a, b$ form an arithmetic sequence and $a, b, 18$ form a geometric sequence. Find a, b .

Classified Practice

Geometric Sequences

Q110 Challenge

A planning model gives $-5, a, b$ as an arithmetic sequence, while $a, b, 45$ form a geometric sequence. Find a, b .

Classified Practice

Geometric Sequences

Q111 Written

Find the sum S of a geometric sequence with first term 4, common ratio 3, and 5 terms.

Q112 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 5 terms.

Classified Practice

Geometric Sequences

Q113 Written

Find the sum S of a geometric sequence with first term 5, common ratio 3, and 4 terms.

Q114 Written

Find the sum S of a geometric sequence with first term 3, common ratio 2, and 10 terms.

Classified Practice

Geometric Sequences

Q115 MCQ

For a geometric sequence with $a = 4$, $r = 3$, and $n = 4$, calculate S_n :

Choose the correct option.

A 161

B 159

C 162

D 160

Q116 MCQ

For a geometric sequence with $a = 5$, $r = 2$, and $n = 6$, calculate S_n :

Choose the correct option.

A 316

B 314

C 315

D 317

Classified Practice

Geometric Sequences

Q117 MCQ

For a geometric sequence with $a = 2$, $r = 4$, and $n = 5$, calculate S_n :

Choose the correct option.

A 683

B 682

C 681

D 684

Q118 MCQ**For a geometric sequence with $a = 3$, $r = 5$, and $n = 4$, calculate S_n :****Choose the correct option.****A** 468**B** 469**C** 467**D** 470

Classified Practice

Geometric Sequences

Q119 MCQ

For a geometric sequence with $a = 7$, $r = 2$, and $n = 5$, calculate S_n :

Choose the correct option.

A 218

B 216

C 219

D 217

Q120 MCQ**For a geometric sequence with $a = 6$, $r = 3$, and $n = 3$, calculate S_n :****Choose the correct option.****A** 79**B** 77**C** 78**D** 80

Classified Practice

Geometric Sequences

Q121 Written

Find the sum S of a geometric sequence with first term 8, common ratio $\frac{1}{2}$, and 6 terms.

Q122 Written

Find the sum S of a geometric sequence with first term 6, common ratio $\frac{1}{2}$, and 6 terms.

Classified Practice

Geometric Sequences

Q123 Written

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{3}$, and 3 terms.

Q124 Written

Find the sum S of a geometric sequence with first term 1, common ratio $\frac{1}{2}$, and 8 terms.

Classified Practice

Geometric Sequences

Q125 MCQ

For a geometric sequence with $a = 8$, $r = \frac{1}{2}$, and $n = 4$, calculate S_n :

Choose the correct option.

A 16

B 15

C 14

D 17

Q126 MCQ

For a geometric sequence with $a = 9$, $r = \frac{1}{3}$, and $n = 3$, calculate S_n :

Choose the correct option.

- A 13
- B 14
- C 12
- D 15

Classified Practice

Geometric Sequences

Q127 MCQ

For a geometric sequence with $a = 12$, $r = \frac{1}{2}$, and $n = 5$, calculate S_n :

Choose the correct option.

A $\frac{47}{2}$

B $\frac{23}{95}$

C $\frac{4}{93}$

D $\frac{93}{4}$

Q128 MCQ

For a geometric sequence with $a = 5$, $r = \frac{1}{4}$, and $n = 3$, calculate S_n :

Choose the correct option.

A $\frac{53}{8}$

B $\frac{13}{2}$

C $\frac{105}{16}$

D $\frac{107}{16}$

Classified Practice

Geometric Sequences

Q129 MCQ

For a geometric sequence with $a = 10$, $r = \frac{1}{2}$, and $n = 6$, calculate S_n :

Choose the correct option.

A $\frac{79}{4}$

B $\frac{315}{16}$

C $\frac{157}{8}$

D $\frac{317}{16}$

Q130 MCQ

For a geometric sequence with $a = 4$, $r = \frac{1}{3}$, and $n = 4$, calculate S_n :

Choose the correct option.

- A $\frac{160}{27}$
- B $\frac{161}{27}$
- C $\frac{53}{9}$
- D 6

Classified Practice

Geometric Sequences

Q131 Written

Find the sum S of a geometric sequence with first term 6, common ratio -2 , and 4 terms.

Q132 Written

Find the sum S of a geometric sequence with first term 1, common ratio -3 , and 4 terms.

Classified Practice

Geometric Sequences

Q133 Written

Find the sum S of a geometric sequence with first term 2, common ratio -3 , and 6 terms.

Q134 MCQ**For a geometric sequence with $a = 4$, $r = -2$, and $n = 5$, calculate S_n :****Choose the correct option.****A** 45**B** 43**C** 46**D** 44

Classified Practice

Geometric Sequences

Q135 MCQ

For a geometric sequence with $a = 3$, $r = -3$, and $n = 4$, calculate S_n :

Choose the correct option.

A -59

B -61

C -60

D -58

Q136 MCQ

For a geometric sequence with $a = 2$, $r = -1$, and $n = 6$, calculate S_n :

Choose the correct option.

A 1

B 0

C -1

D 2

Classified Practice

Geometric Sequences

Q137 MCQ

For a geometric sequence with $a = 5$, $r = -2$, and $n = 6$, calculate S_n :

Choose the correct option.

A -105 B -104 C -106 D -103

Q138 MCQ**For a geometric sequence with $a = 7$, $r = -1$, and $n = 5$, calculate S_n :****Choose the correct option.****A** 8**B** 6**C** 9**D** 7

Classified Practice

Geometric Sequences

Q139 MCQ

For a geometric sequence with $a = 6$, $r = -3$, and $n = 3$, calculate S_n :

Choose the correct option.

A 43

B 41

C 42

D 44

Q140 Written

Find S_n from the first term to the n -th term when the first term is 3 and the common ratio is -2 .

Classified Practice

Geometric Sequences

Q141 **Written**

Find S_n from the first term to the n -th term when the first term is 7 and the common ratio is 2.

Q142 Written

Find S_n from the first term to the n -th term when the first term is 2 and the common ratio is $\frac{1}{3}$.

Classified Practice

Geometric Sequences

Q143 **Written**

Find S_n from the first term to the n -th term when the first term is 4 and the common ratio is 2.

Q144 MCQ

For the geometric sequence with $a = 3, r = 2$, find the sum S_n from the first term to the n -th term:

Choose the correct option.

A $3(2^{n-1} - 1)$

B $6(2^n - 1)$

C $3(2^n - 1)$

D $3(1 - 2^n)$

Classified Practice

Geometric Sequences

Q145 MCQ

For the geometric sequence with $a = 5, r = \frac{1}{2}$, find the sum S_n from the first term to the n -th term:

Choose the correct option.

A $5\{1 - (\frac{1}{2})^n\}$

B $10\{1 - (\frac{1}{2})^{n-1}\}$

C $10\{1 - (\frac{1}{2})^n\}$

D $10\{1 - (\frac{1}{2})^n\}$

Q146 MCQ

For the geometric sequence with $a = 4, r = -2$, find the sum S_n from the first term to the n -th term:

Choose the correct option.

A $\frac{4\{1 - (-2)^n\}}{3}$

B $\frac{4\{1 - (-2)^{n-1}\}}{3}$

C $\frac{4\{1 + (-2)^n\}}{3}$

D $\frac{4\{1 - 2^n\}}{3}$

Classified Practice

Geometric Sequences

Q147 **Written****Find S_n from the first term to the n -th term of 2, 6, 18, 54,**

Q148 Written

Find S_n from the first term to the n -th term of 2, -6 , 18, -54 ,

Classified Practice

Geometric Sequences

Q149 **Written****Find S_n from the first term to the n -th term of $8, -4, 2, -1, \dots$**

Q150 Written

Find S_n from the first term to the n -th term of $1, -3, 9, -27, \dots$

Classified Practice

Geometric Sequences

Q151 Written

Find S_n from the first term to the n -th term when the first term is 16 and the common ratio is $-\frac{1}{2}$.

Q152 Written

Find S_n from the first term to the n -th term of $1, -2, 4, -8, \dots$

Classified Practice

Geometric Sequences

Q153 MCQ

A culture grows by 3, 9, 27, 81 units. Find the four-stage total:

Choose the correct option.

A 120

B 121

C 119

D 122

Classified Practice

Geometric Sequences

Q154 Challenge

A filtration scale is geometric: $3, 1, \frac{1}{3}, \frac{1}{9}, \dots$ mg/L removed per stage. Calculate the total removed in 4 stages.

Classified Practice

Geometric Sequences

Q155 MCQ

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is 28, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (2, 2)$

B $(a, r) = (1, 3)$

C $(a, r) = \left(\frac{10}{3}, -2\right)$

D $(a, r) = (1, 2)$ or $(a, r) = \left(\frac{7}{3}, -2\right)$

Classified Practice

Geometric Sequences

Q156 MCQ

Given $S_3 = 26$ and the sum from the 3rd to the 5th terms is 234, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (3, 3)$

B $(a, r) = (2, 4)$

C $(a, r) = (2, 3)$ or $(a, r) = \left(\frac{26}{7}, -3\right)$

D $(a, r) = \left(\frac{33}{7}, -3\right)$

Classified Practice

Geometric Sequences

Q157 MCQ

Given $S_3 = 21$ and the sum from the 3rd to the 5th terms is 84, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (4, 2)$

B $(a, r) = (3, 2)$ or $(a, r) = (7, -2)$

C $(a, r) = (3, 3)$

D $(a, r) = (8, -2)$

Classified Practice

Geometric Sequences

Q158 MCQ

Given $S_3 = 28$ and the sum from the 3rd to the 5th terms is 112, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (4, 2)$ or $(a, r) = \left(\frac{28}{3}, -2\right)$

B $(a, r) = (5, 2)$

C $(a, r) = (4, 3)$

D $(a, r) = \left(\frac{31}{3}, -2\right)$

Classified Practice

Geometric Sequences

Q159 MCQ

Given $S_3 = 13$ and the sum from the 3rd to the 5th terms is 117, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (2, 3)$

B $(a, r) = (1, 4)$

C $(a, r) = \left(\frac{20}{7}, -3\right)$

D $(a, r) = (1, 3)$ or $(a, r) = \left(\frac{13}{7}, -3\right)$

Classified Practice

Geometric Sequences

Q160 MCQ

Given $S_3 = 35$ and the sum from the 3rd to the 5th terms is 140, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (6, 2)$

B $(a, r) = (5, 3)$

C $(a, r) = (5, 2)$ or $(a, r) = \left(\frac{35}{3}, -2\right)$

D $(a, r) = \left(\frac{38}{3}, -2\right)$

Classified Practice

Geometric Sequences

Q161 MCQ

Given $S_3 = \frac{19}{2}$ and the sum from the 3rd to the 5th terms is $\frac{171}{8}$, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = \left(3, \frac{3}{2}\right)$

B $(a, r) = \left(2, \frac{5}{2}\right)$

C $(a, r) = \left(\frac{45}{7}, -\frac{3}{2}\right)$

D $(a, r) = \left(2, \frac{3}{2}\right)$ or $(a, r) = \left(\frac{38}{7}, -\frac{3}{2}\right)$

Classified Practice

Geometric Sequences

Q162 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 4th to the 6th term is -72 .

Classified Practice

Geometric Sequences

Q163 MCQ

Given $S_3 = 7$ and the sum from the 3rd to the 5th terms is $\frac{7}{4}$, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = \left(5, \frac{1}{2}\right)$

B $(a, r) = \left(4, \frac{3}{2}\right)$

C $(a, r) = \left(4, \frac{1}{2}\right)$ or $(a, r) = \left(\frac{28}{3}, -\frac{1}{2}\right)$

D $(a, r) = \left(\frac{31}{3}, -\frac{1}{2}\right)$

Classified Practice

Geometric Sequences

Q164 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 35$ and the sum from the 3rd to the 5th term is 140.

Q165 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and the sum from the 3rd to the 5th term is 84.

Classified Practice

Geometric Sequences

Q166 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 9$ and the sum from the 3rd to the 5th term is 36.

Q167 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 78$ and the sum from the 3rd to the 5th term is 702.

Classified Practice

Geometric Sequences

Q168 MCQ

Given $S_3 = 6$ and the sum from the 3rd to the 5th terms is 24, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (3, -2)$

B $(a, r) = (2, -2)$ or $(a, r) = \left(\frac{6}{7}, 2\right)$

C $(a, r) = (2, -1)$

D $(a, r) = \left(\frac{13}{7}, 2\right)$

Classified Practice

Geometric Sequences

Q169 MCQ

Given $S_3 = 9$ and the sum from the 3rd to the 5th terms is 36, find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (3, -2)$ or $(a, r) = \left(\frac{9}{7}, 2\right)$

B $(a, r) = (4, -2)$

C $(a, r) = (3, -1)$

D $(a, r) = \left(\frac{16}{7}, 2\right)$

Classified Practice

Geometric Sequences

Q170 **Written**

Find the first term and common ratio of a geometric sequence if the 2nd term is 3 and $S_3 = 13$.

Q171 Written

Find the first term and common ratio of a geometric sequence if the 2nd term is 6, and $S_3 = 21$.

Classified Practice

Geometric Sequences

Q172 MCQ

The second term is 6 and $S_3 = 21$. Find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (4, 2)$

B $(a, r) = (3, 3)$

C $(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$

D $(a, r) = \left(13, \frac{1}{2}\right)$

Classified Practice

Geometric Sequences

Q173 MCQ

The second term is 6 and $S_3 = 26$. Find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (3, 3)$

B $(a, r) = (2, 4)$

C $(a, r) = \left(19, \frac{1}{3}\right)$

D $(a, r) = (2, 3)$ or $(a, r) = \left(18, \frac{1}{3}\right)$

Classified Practice

Geometric Sequences

Q174 MCQ

The second term is 8 and $S_3 = 28$. Find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (5, 2)$

B $(a, r) = (4, 2)$ or $(a, r) = \left(16, \frac{1}{2}\right)$

C $(a, r) = (4, 3)$

D $(a, r) = \left(17, \frac{1}{2}\right)$

Classified Practice

Geometric Sequences

Q175 MCQ

The second term is 3 and $S_3 = 13$. Find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = (2, 3)$

B $(a, r) = (1, 4)$

C $(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$

D $(a, r) = \left(10, \frac{1}{3}\right)$

Classified Practice

Geometric Sequences

Q176 MCQ

The second term is $\frac{5}{2}$ and $S_3 = \frac{35}{4}$. Find all real pairs (a, r) :

Choose the correct option.

A $(a, r) = \left(6, \frac{1}{2}\right)$

B $(a, r) = \left(5, \frac{3}{2}\right)$

C $(a, r) = \left(\frac{9}{4}, 2\right)$

D $(a, r) = \left(5, \frac{1}{2}\right)$ or $(a, r) = \left(\frac{5}{4}, 2\right)$

Classified Practice

Geometric Sequences

Q177 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 4 and $S_3 = 7$.

Q178 Written

Find the first term and common ratio of a geometric sequence if the 3rd term is 12, and $S_3 = 9$.

Classified Practice

Geometric Sequences

Q179 **Written**

Find the first term and common ratio of a geometric sequence if the 3rd term is 20, and $S_3 = 35$.

Q180 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 21$ and $S_6 = 189$.

Classified Practice

Geometric Sequences

Q181 **Written**

Find the first term and common ratio of a geometric sequence if $S_3 = 26$ and $S_6 = 728$.

Q182 Written

Find the first term and common ratio of a geometric sequence if $S_3 = 65$ and $S_6 = 1820$.

Classified Practice

Geometric Sequences

Q183 **Written**

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is 3, and $S_4 = 51$.

Q184 Written

Find the first term and common ratio of a geometric sequence if the sum of the first and second terms is -8 , and $S_4 = -80$.

Classified Practice

Geometric Sequences

Q185 MCQ

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} :

Choose the correct option.

A 537

B 487

C 562

D 512

Classified Practice

Geometric Sequences

Q186 Challenge

If S_n is the sum of the first n terms of a geometric sequence, $S_8 = 255$, $S_6 = 63$, and $a_8 - a_7 = 64$, find a_{10} .

Classified Practice

Geometric Sequences

Quiz MCQ 1 [Concept Quiz](#) · MCQ

Select the correct option: For $a = 3, r = 2$, find a_6 :

Choose the correct option.

A 48

B 96

C 192

D 64

Quiz MCQ 2 Concept Quiz · MCQ

Select the correct option: If 2, x , 8 are consecutive geometric terms, find x :

Choose the correct option.

- A 4
- B -4
- C ± 4
- D 16

Classified Practice

Geometric Sequences

Quiz MCQ 3 Concept Quiz · MCQ

For 1, 3, 9, ..., find S_n :

Choose the correct option.

A $\frac{3^{n-1} - 1}{2}$

B $\frac{3^n - 1}{2}$

C $\frac{3^n + 1}{2}$

D $\frac{3^n - 1}{3}$

Quiz MCQ 4 Concept Quiz · MCQ

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} :

Choose the correct option.

A 528

B 496

C 544

D 512

Classified Practice

Geometric Sequences

Quiz WRITTEN 5 Concept Quiz · Written**Find the general term of $-4, 8, -16, 32, \dots$**

Quiz WRITTEN 6 Concept Quiz · Written

If 1, x , 25 are consecutive geometric terms, find x .

Classified Practice

Geometric Sequences

Quiz WRITTEN 7 Concept Quiz · Written

A four-stage process has outputs 5, 15, 45, 135. Find the total output.

Quiz WRITTEN 8 Concept Quiz · Written

A geometric sequence has $S_6 = 63$, $S_7 = 127$, and $S_8 = 255$. Find a_{10} .

Answer key

Use the question number shown on each page.

Q001 $2 \times 3^{n-1}$

Q002 $3 \times 4^{n-1}$

Q003 $5 \times 2^{n-1}$

Q004 $2 \times 5^{n-1}$

Q005 $7 \times 4^{n-1}$

Q006 A

Q007 A

Q008 A

Q009 A

Q010 A

Q011 A

Q012 $5(-2)^{n-1}$

Q013 $2(-3)^{n-1}$

Q014 $3(-5)^{n-1}$

Q015 $4(-2)^{n-1}$

Q016 A

Q017 A

Q018 A

Q019 A

Q020 A

Q021 $8\left(-\frac{1}{3}\right)^{n-1}$

Q022 $4\left(\frac{1}{3}\right)^{n-1}$

Q023 $3\left(-\frac{1}{2}\right)^{n-1}$

Q024 $6\left(\frac{1}{2}\right)^{n-1}$

Q025 A

Q026 A

Q027 A

Q028 A

Q029 A

Q030 $5 \times 2^{n-1}$

Q031 $2 \times 3^{n-1}$

Q032 $3 \times 2^{n-1}$

Q033 A

Q034 A

Q035 A

Q036 A

Q037 A

Q038 $2(-3)^{n-1}$

Q039 $5(-2)^{n-1}$

Q040 A

Q041 A

Q042 A

Q043 A

Q044 $-4\left(-\frac{3}{2}\right)^{n-1}$

Q045 $-8\left(-\frac{3}{2}\right)^{n-1}$

Q046 $2\left(\frac{1}{3}\right)^{n-1}$

Q047 $48\left(\frac{1}{2}\right)^{n-1}$

Q048 $\frac{1}{6}\left(\frac{3}{2}\right)^{n-1}$

Q049 $8(2\sqrt{2})^{n-1}$

Q050 $3\left(-\frac{5}{2}\right)^{n-1}$

Q051 $9\left(-\frac{\sqrt{3}}{3}\right)^{n-1}$

Q052 $5\left(-\frac{2}{3}\right)^{n-1}$

Q053 A

Q054 A

Q055 A

Q056 A

Q057 A

Q058 A

Q059 A

Q060 $r = -\frac{1}{3}, A = 54, B = -2$

Q061 A

Q062 $r = \frac{1}{2}, A = 48, B = 6$

Q063 $r = -3, A = 1, B = -27$

Q064 $r = \frac{1}{3}, A = 1, B = \frac{1}{27}$

Q065 $r = -\frac{1}{2}, A = \frac{1}{2}, B = -\frac{1}{16}$

Q066 $a_7 = 648$

Q067 $a_n = 3(2)^{n-1}$ or $-3(-2)^{n-1}$

Q068 $a_n = 2(3)^{n-1}$ or $2(-3)^{n-1}$

Q069 $a_n = 2(3)^{n-1}$ or $-2(-3)^{n-1}$

Q070 $a_n = 3(3)^{n-1}$ or $3(-3)^{n-1}$

Q071 $a_n = 7(2)^{n-1}$

Q072 $a_n = 3(-2)^{n-1}$

Q073 $a_n = 3(-2)^{n-1}$

Q074 $a_n = 3(2)^{n-1}$ or $3(-2)^{n-1}$

Q075 $a_n = -7(2)^{n-1}$ or $-7(-2)^{n-1}$

Q076 $a_n = -2(-2)^{n-1}$

Q077 $a_n = -5(-2)^{n-1}$

Q078 $a_n = \frac{3}{4}(4)^{n-1}$ or $-\frac{3}{4}(-4)^{n-1}$

Q079 $a_n = 4\left(\frac{3}{2}\right)^{n-1}$ or $4\left(-\frac{3}{2}\right)^{n-1}$

Q080 $a_n = \frac{2}{243}(3)^{n-1}$ or $-\frac{2}{243}(-3)^{n-1}$

Q081 A

Q082 A

Q083 A

Q084 A

Q085 A

Q086 A

Q087 A

Q088 A

Q089 A

Q090 A

Q091 A

Q092 A

Q093 A

Q094 A

Q095 A	Q122 $\frac{189}{16}$	Q148 $\frac{1 - (-3)^n}{2}$	Q171 $(a, r) = (3, 2)$ or $(a, r) = \left(12, \frac{1}{2}\right)$
Q096 A	Q123 $\frac{13}{9}$	Q149 $\frac{16}{3}\{1 - (-\frac{1}{2})^n\}$	Q172 C
Q097 $x = \pm 4$	Q124 $\frac{255}{128}$	Q150 $\frac{1 - (-3)^n}{4}$	Q173 D
Q098 $x = \pm 5$	Q125 B	Q151 $\frac{32}{3}\{1 - (-\frac{1}{2})^n\}$	Q174 B
Q099 $x = \pm 2$	Q126 A	Q152 $\frac{1 - (-2)^n}{3}$	Q175 C
Q100 $x = \pm 2\sqrt{3}$	Q127 D	Q153 A	Q176 D
Q101 A	Q128 C	Q154 $\frac{40}{9}$	Q177 $(a, r) = (1, 2)$ or $(a, r) = \left(9, -\frac{2}{3}\right)$
Q102 A	Q129 B	Q155 D	Q178 $(a, r) = (3, -2)$
Q103 A	Q130 A	Q156 C	Q179 $(a, r) = (5, 2)$ or $(a, r) = \left(45, -\frac{2}{3}\right)$
Q104 A	Q131 -30	Q157 B	Q180 $(a, r) = (3, 2)$
Q105 A	Q132 -20	Q158 A	Q181 $(a, r) = (2, 3)$
Q106 A	Q133 -364	Q159 D	Q182 $(a, r) = (5, 3)$
Q107 $(a, b) = (9, 12)$ or $(1, -4)$	Q134 D	Q160 C	Q183 $(a, r) = \left(\frac{3}{5}, 4\right)$ or $(a, r) = (-1, -4)$
Q108 $(a, b) = (2, 2)$ or $(-4, -1)$	Q135 C	Q161 D	Q184 $(a, r) = (-2, 3)$ or $(a, r) = (4, -3)$
Q109 $(a, b) = (2, 6)$ or $(\frac{1}{2}, 3)$	Q136 B	Q162 $(a, r) = (3, -2)$	Q185 D
Q110 $(a, b) = (5, 15)$ or $(\frac{5}{4}, \frac{15}{2})$	Q137 A	Q163 C	Q186 $a_7 = 64, a_8 = 128, r = 2, a_{10} = 512$
Q111 484	Q138 D	Q164 $(a, r) = (5, 2)$ or $(a, r) = \left(\frac{35}{3}, -2\right)$	Quiz MCQ 1 B
Q112 93	Q139 C	Q165 $(a, r) = (3, 2)$ or $(a, r) = (7, -2)$	Quiz MCQ 2 C
Q113 200	Q140 $1 - (-2)^n$	Q166 $(a, r) = \left(\frac{9}{7}, 2\right)$ or $(a, r) = (3, -2)$	Quiz MCQ 3 B
Q114 3069	Q141 $7(2^n - 1)$	Q167 $(a, r) = (6, 3)$ or $(a, r) = \left(\frac{78}{7}, -3\right)$	Quiz MCQ 4 D
Q115 D	Q142 $3\{1 - (\frac{1}{3})^n\}$	Q168 B	Quiz WRITTEN 5 $-4(-2)^{n-1}$
Q116 C	Q143 $4(2^n - 1)$	Q169 A	Quiz WRITTEN 6 $x = \pm 5$
Q117 B	Q144 C	Q170 $(a, r) = (1, 3)$ or $(a, r) = \left(9, \frac{1}{3}\right)$	Quiz WRITTEN 7 200
Q118 A	Q145 D		Quiz WRITTEN 8 512
Q119 D	Q146 A		
Q120 C	Q147 $3^n - 1$		
Q121 $\frac{63}{4}$			